

DRY-RUN GUIDE | 08_lcm.cpp
Source: cpp\2nd_belt\08_lcm.cpp

CORE BEHAVIOR

Returns zero if either input is zero; otherwise computes $|(a/\gcd(|a|,|b|))*b|$ using Euclid's algorithm.

REPRESENTATIVE INPUT / CALL

Input: '4 6'.

STEP TRACE

$\gcd(4,6)=2$; $(4/2)*6=12$; absolute value remains 12.

EXPECTED OUTPUT / RETURN

'12'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Write a program to calculate and print the Least Common Multiple (LCM)
004 | of two numbers. If either number is zero, the LCM of the two numbers is
005 | also zero.
006 |
007 | Input Format:
008 | The input consists of two integers A and B.
009 |
010 | Output Format:
011 | The output consists of the LCM of the two numbers.
012 |
013 | Example:
014 | Input: 4 6
015 | Output: 12
016 | */
017 |
018 | #include <iostream>
019 | #include <cstdlib>
020 | using namespace std;
021 |
022 | long long gcd(long long a, long long b) {
023 |     while (b != 0) {
024 |         long long r = a % b;
025 |         a = b;
026 |         b = r;
027 |     }
028 |     return a;
029 | }
030 |
031 | int main() {
032 |     long long a, b;
033 |     cin >> a >> b;
034 |
035 |     if (a == 0 || b == 0) {
036 |         cout << 0;
037 |     } else {
```

```
038 |         cout << llabs((a / gcd(llabs(a), llabs(b))) * b);
039 |     }
040 |
041 |     return 0;
042 | }
043 |
```